Government Engineering College, Gandhinagar

"Problem Based Learning"

Under subject of

Python for Data Science

Branch: Computer Engineering

Class / Division: 5th CE B

Submitted by:

1.	**Darshan Rajgor**	(240130107115)
2.	**Prince Suthar**	(240130107139)
3.	**Pranay Prajapati**	(240130107111)
4.	**Smit Sutariya**	(240130107138)
5.	**Meet Raval**	(240130107118)

Prof. V.R. Patel
(Subject Cordinator)

Dr. D. A. Parikh
(HOD OF CE)

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Resume Skill Gap Analyzer using NLP

Project Description and Objectives :

An NLP-based system designed to analyze uploaded student resumes against target job descriptions to identify key missing skills and bridge the gap between candidate qualifications and employer expectations.

Objectives

- Extract text data efficiently from unstructured resume file formats (PDF/DOCX) and text-based job descriptions.
- Apply Natural Language Processing (NLP) techniques to extract key skills, technical terms, and qualifications.
- Compute similarity scores between candidate profiles and job requirements.
- Provide an automated summary of identified missing skills to help students optimize their technical profiles for specific job roles.

Dataset to Be Used :

Source: Publicly available datasets from Kaggle (e.g., Resume Dataset and Job Descriptions Dataset).

Brief Details:

Data Source	Description
Resume Data	Structured and unstructured text containing candidate profiles across multiple domains (e.g., Data Science, Web Development, Software Engineering) detailing skills, experience, and education.
Job Description Data	Collection of job postings containing required skill sets, role descriptions, responsibilities, and key qualifications.

Methodology Diagram :

1. Data Collection
Gather candidate resumes and job descriptions from Kaggle datasets.

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2. Text Preprocessing
Extract text from PDF/DOCX, normalize text (lowercasing), tokenize, and remove stop words/punctuation.

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3. Feature & Skill Extraction
Apply Named Entity Recognition (NER) and keyword/phrase extraction to identify skills and qualifications.

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4. Vectorization & Matching Engine
Convert text using TF-IDF / Sentence Transformers and compute Cosine Similarity between resume and job description.

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5. Skill Gap Analysis
Compare extracted resume skills against job requirements to identify missing technical keywords.

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6. Output & Visualization
Generate match percentage score, list matched skills, and highlight missing skills for candidate profile optimization.